



Environmental Discrimination in Plant Breeding
Programs]Method for Estimating Genotype Stability, Adaptability, and
Environmental Discrimination in Plant Breeding
Programs

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Abstract:

Breeding programs aim not only to improve crop performance but also to ensure stability and adaptability across diverse environments.

Evaluating genotype resilience requires understanding both general adaptability ability (GAA), the mean trait performance across multiple environments, as well as specific adaptability ability (SAA), the deviation of a genotype in a particular environment. A comprehensive approach for assessing GAA, SAA, and genotype stability, is presented for integrating both linear and nonlinear responses to environmental variation. The method relies on multi-environment trials, partitioning phenotypic variance into genotype, environment, and genotype $\times$ environment interaction components using ANOVA models, and calculating stability indices such as the relative stability and stability value of genotypes (SVG). This framework allows breeders to select genotypes with optimal performance while maintaining stability, balancing trade-offs between high mean trait values and environmental sensitivity.

Furthermore, it provides insights into environmental differentiation and the compensatory or destabilizing effects of genotype $\times$ environment interactions. By combining theoretical and practical aspects of adaptive breeding, this approach offers a robust tool for guiding selection strategies that enhance both productivity and resilience in crop populations. To compare populations and select appropriate breeding methods across different environments, a technique is introduced to partition the phenotypic variance of a population into GAA and SAA variances.

Keywords: Adaptability ability, Stability value, Genotype $\times$ environment interaction

Mathematics Subject Classification (2010): 62P10, 62J10, 62K10, 92D10.

1 Introduction

Plant breeders are challenged not only with the task of increasing plant performance (productivity), but also with combining performance with resistance to biotic and abiotic environmental conditions. Accordingly, researchers have focused on studying and developing methods to evaluate resilience (stability, general and specific adaptability of genotypes), as well as breeding approaches aimed at optimizing these important plant traits. Various approaches exist for estimating genotype stability. The terms general and specific adaptability (Annicchiarico, 2002), which reflect the overall response of a genotype across all environments and its specific response in a particular environment, have been reported in the literature. Studying genotypes across different environments can also provide information about the environments themselves as a context for breeding. Kilchevskii & Khotilova, (1985), proposed a genetic analysis method based on testing genotypes in multiple environments that allows researchers to determine the general and specific adaptability and stability of genotypes, and to compare environments in terms of their ability to differentiate genotypes. In this approach, the breeding value of a genotype can be estimated depending on the breeding objectives, and the breeding process can be guided according to adaptability levels. This method also enables the partitioning of a population's phenotypic variance into general and specific adaptability variances, allowing comparisons among populations and informed selection of breeding methods.

ability]General and specific adaptability
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Adaptability is defined as the ability of a genotype to maintain its specific phenotypic trait under a particular environmental condition. General adaptability ability (GAA) of a genotype represents the mean trait value across different environmental conditions, while specific adaptability ability (SAA) represents the deviation from GAA in a specific environment. The method for evaluating GAA and SAA is based on testing a population of n genotypes across m environments. The number of replications is k . Accordingly; x_{ikr} is the phenotypic value of genotype i in replication r in environment k ; u is the overall mean of all phenotypic values; v_i is the effect of genotype i ; d_k is the effect of environment k ; $(vd)_{ik}$ is the interaction effect of genotype i with environment k ; e_{ikr} is the random error associated with phenotype ikr .

To determine the significance of genotype, environment, and genotype $\times$ environment interaction effects on the phenotypic variation of the population, a two-way analysis of variance (ANOVA) is used. Interpretation of the ANOVA results depends on the underlying

assumptions of the study. If genotypes and environments are non-random samples, their effects are considered fixed, and information can be obtained about specific genotypes and environments. In this case, Model 1 is used, in which all effect values, except for the independent variable e_{ikr} are considered fixed. The variable e_{ikr} is assumed to be normally distributed with mean zero and variance σ_e^2 . The F-test is used to determine the significance of genotype, environment, and interaction effects. In this test, the corresponding mean squares are compared to the mean square of the random deviations. If genotypes are a random sample from the population and environments are a random sample from all possible environments, then Model 2 is used. In this case, it is possible to obtain information about the population parameters of genotypes for a specific set of environments. In Model 2, only the overall mean μ is considered fixed, while the other effects are independently and normally distributed with mean zero and variance σ^2 . In this case, to test hypotheses regarding genotypes and environments, the mean square of genotype $\times$ environment interaction is used, and to estimate the genotype $\times$ environment interaction effect, the mean square of random deviations is applied.

2 Description of method

Regarding accept the assumptions of Model 1, the effect of the general adaptability ability (GAA) of genotype i is equal to $\mu + v_i$. The deviation from the sum $\mu + v_i$ represents the specific adaptability ability effect (SAA_{ik}) of genotype i in environment k . This effect consists of a linear component (the effect of environment d_k) and a nonlinear component (the interaction effect $(vd)_{ik}$). To determine the effects of GAA_i and SAA_{ik} , the mean phenotypic values across replications are calculated. In this way, the results of testing n genotypes across m environments can be arranged in an $n \times m$ table, where: x_i is the sum of all phenotypic values for genotype i , x_k is the total phenotypic sum in environment k , $X_{..}$ is the grand total of all phenotypic values, x_{ik} is the phenotypic value of genotype i in environment k . The GAA and SAA effects are calculated using the following formulas (Fig. 1):

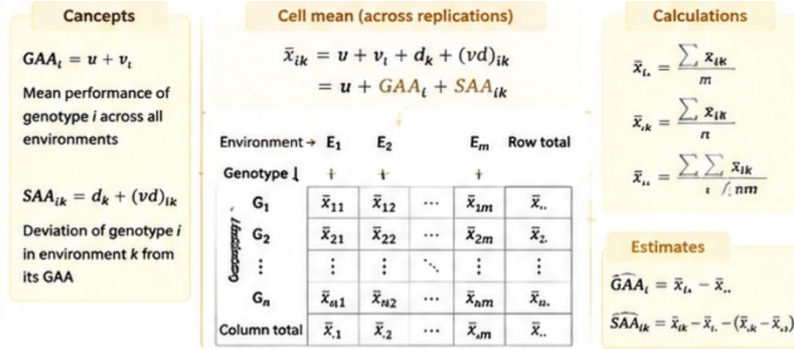
$$x_{i.} = \sum_k x_{ik} = x_{i1} + x_{i2} + \dots + x_{im}$$

$$x_{nk} = \sum_i x_{ik} = x_{1k} + x_{2k} + \dots + x_{nk}$$

$$x_{ik} = u + v_i + d_k + (vd)_{ik}$$

$$x_{ik} = u + GAA_i + SAA_{ik}$$

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Figure 1. Schematic illustration of stability evaluation

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The variance of random deviations for the mean phenotypic value in a specific environment x_{ik} is computed as:

$$var_{x_{ik}} = \frac{\sigma_e^2}{c} = \sigma^2$$

The variance of the difference between the means of two genotypes is determined as:

$$var_{(x_{ik}/x_{nm})} = 2\sigma^2$$

To obtain the standard error of the differences between effects, the square root of the corresponding variance must be taken. The least significant difference (LSD) is then obtained by multiplying the standard error by a constant magnitude. Comparison of genotypes in terms of general adaptability can be performed by comparing GAA_i values. To estimate the interaction strength of genotype i with environments, the genotype $\times$ environment interaction variance can be used. To measure the stability of genotype i , it is recommended to use the variance of SAA_i . Therefore, stability can be defined as the capacity of a genotype to maintain its phenotypic expression under various environmental conditions through organized mechanisms.

3 Stability analysis

Stabilizing selection provides the evolutionary basis for developing organized mechanisms that ensure the phenotypic stability of genotypes. In this way, stability in the expression of one trait can be combined with flexibility in the expression of other traits. The proposed method for estimating stability is based on combining the linear and nonlinear components of the genotype's response to the environment. This approach differs from other established methods like regression model of Finlay and Wilkinson (1963), in which stability is measured using a linear response; ecovalence in which stability is estimated via a nonlinear response; and regression models of Eberhart & Russell (1966), in which parameters for both linear and nonlinear genotype responses are determined.

The concept of relative genotype stability is defined as follows:

$$s_{gi} = \frac{\sigma_{GAA_i}}{u + SAA_i} \times 100$$

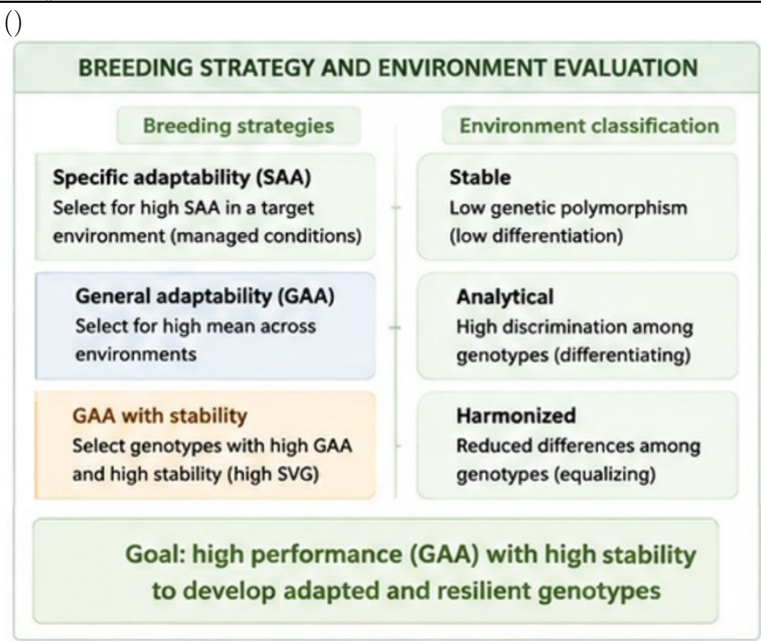
This index allows comparison of experimental results obtained across different sets of crops, genotypes, environments, and traits under study. Essentially, relative stability is analogous to the coefficient of variation when studying a genotype across multiple environments. The ratio l_{gi} can serve as a nonlinear index of genotype i 's response to the environment:

$$l_{gi} = \frac{\sigma_{GE}^2}{\sigma_{SA}^2}$$

If $l_{gi} \rightarrow 1$, the genotype exhibits mostly nonlinear responses across most environments, while if $l_{gi} \rightarrow 0$, the genotype's response to the environment is predominantly linear. The response of a genotype to improved environmental conditions can be quantified using the regression coefficient of the genotype across environments. There is considerable disagreement among breeders regarding the interpretation of genotype stability. Consequently, Eberhart and Russell (1966), suggested that an ideal genotype should have: a high mean trait value, a regression coefficient close to 1, and the smallest deviation from the regression line. However, the work of Mulder et al. (2007), demonstrated that through breeding, any combination of mean trait value and environmental sensitivity can be achieved. The most acceptable perspective is that of Finlay and Wilkinson (1963), regression model which is defined an optimal genotype as one that possesses, as high general adaptability, maximum performance under favorable environments, and maximum stability. According to Martínez-Peña et al. (2023), a genotype with moderate but stable yield has higher economic value than a genotype with high potential but highly variable yield. Similarly, Bailey-Serres et al. (2019) reported that in crop production, only 30-50% of the potential productivity of dense genotypes is realized, primarily due to

insufficient environmental stability.
The final evaluation of a breeding genotype depends on the objectives of adaptive breeding (Fig. 2). The main directions for adaptive breeding are; breeding genotypes for specific adaptability (SAA) in a particular environment. Such selection is only desirable under controlled and managed environmental conditions (e.g., greenhouse, irrigated field). At the same time, there is a risk of erosion of the genetic basis of general viability and reduced genotype stability. Breeding genotypes for general adaptability (GAA) across multiple environments, selection is based on the mean phenotypic value across all environments. According to Omel'yanyuk (2013), this approach achieves maximum trait enhancement compared to selection in favorable or unfavorable environments, while providing moderate environmental sensitivity. The latter is not controlled during the breeding process.

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Figure 2. Flowchart of breeding and selection strategy

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stability]Breeding for GAA while considering stability
The theoretical basis of this adaptive breeding approach is concerning the inseparable forms of natural selection: directional and stabilizing. The first form alters the structure and performance of organisms, while

the second stabilizes morphogenesis, leading to automated ontogeny and the development of regulatory mechanisms. The effectiveness of stabilizing selection in plants has been demonstrated in several studies (Cresswell, 1998; Rolhauser & Pucheta, 2017). However, stabilizing selection alone cannot create new traits; it is only important for preserving valuable population characteristics and increasing resistance to environmental fluctuations. Breeding for GAA with consideration of stability requires a specific selection criterion, which allows the combination of performance and stability in the genotype. Kilchevskii & Khotilova, (1985), proposed the stability value genotype (SVG) as:

$$SVG_{gi} = \frac{\sqrt{\left(\sum (SAA_{gi})^2\right) - \left(\frac{MSE}{df}\right)}}{u + SAA_i} \times 100$$

In practice, applying the usual selection criterion $p=2$ is unlikely, because for large values of SAA_i , negative SVG values may occur, and some fertility genes could be lost. As a first step toward achieving an optimal balance between performance and stability in breeding, we suggest using the population mean SVG equal to 0.5 grand mean. This facilitates establishing the relationship between the average relative stability of genotypes and the value of p . In this approach, at the lowest value, selection prioritizes stability, while at the highest value, selection favors performance more intensely. Environmental evaluation is also crucial in breeding. Environments can be classified into three types; stable, in which population polymorphism does not appear; analytical, which helps identify different biotypes; harmonized, where the viability of different biotypes is constrained and differences among them are equalized. The above analysis method allows not only the evaluation of genotype adaptability parameters, but also defining the environment as a breeding context.

When interpreting SAA and GAA variances, a question arises: how informative are they, and can they replace genotype $\times$ environment interaction variance? Some researchers believe that genotype $\times$ environment interaction variance can serve as a measure of genotype stability. However, statistical evaluation does not reflect all biological realities of genotype $\times$ environment interaction, including increase or decrease in genotype (or environment) effects. If the signs of genotype (environment) effects and the interaction do not match, the variance of environmental differentiation or genotype stability (compensation effect) decreases. If the signs are aligned, the corresponding variance increases (destabilizing effect). Thus, even if genotype $\times$ environment interaction variance is identical, the variance of genotype stability and the variance of environmental differentiation may differ. The practical importance of the compensation/destabilization effect lies in the unpredictability of genotype stability or environmental differentiation when using

²_{GE}, as a criterion for selecting genotypes or environments. Therefore, SAA and GAA variances provide more objective information on genotype stability and environmental differentiation compared to genotype $\times$ environment interaction variance.

4 Conclusion

This study introduces an integrated framework for evaluating genotype adaptability and stability based on general adaptability ability (GAA) and specific adaptability ability (SAA). By partitioning phenotypic variation into general and environment-specific components, the approach provides a more informative assessment of genotype performance across environments than conventional GE interaction analysis alone. The use of GAA, SAA, and stability indices enables simultaneous selection for high yield and stability, supporting more effective breeding decisions.

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